The composite has speed v_1, momentum p_1, and energy E_1.

$$p_1 = \frac{m_1 v_1}{\sqrt{1 - v_1^2/c^2}} \tag{10}$$

$$E_1 = \frac{m_1 c^2}{\sqrt{1 - v_1^2/c^2}} \tag{11}$$

Momentum and energy are conserved in combination:

$$p_1 = p \tag{12}$$

$$E_1 = E + m_2 c^2 \tag{13}$$

Given $m_2 = 4m_0$, these equations give:

$$v_1 = \frac{v}{1+4\sqrt{1-v^2/c^2}} \tag{14}$$

Substitute (5) into (14):

$$v_1 = 1.43\times10^8 \ \text{m·s}^{-1} \tag{15}$$

From (9), (11), and (13):

$$m_1 = \frac{m_0\sqrt{1-v_1^2/c^2}}{\sqrt{1-v^2/c^2}} + m_2\sqrt{1-v_1^2/c^2} \tag{16}$$

Using (5), (15), and the data:

$$m_1 = 46.4\times10^{-27}\,\text{kg} \tag{17}$$